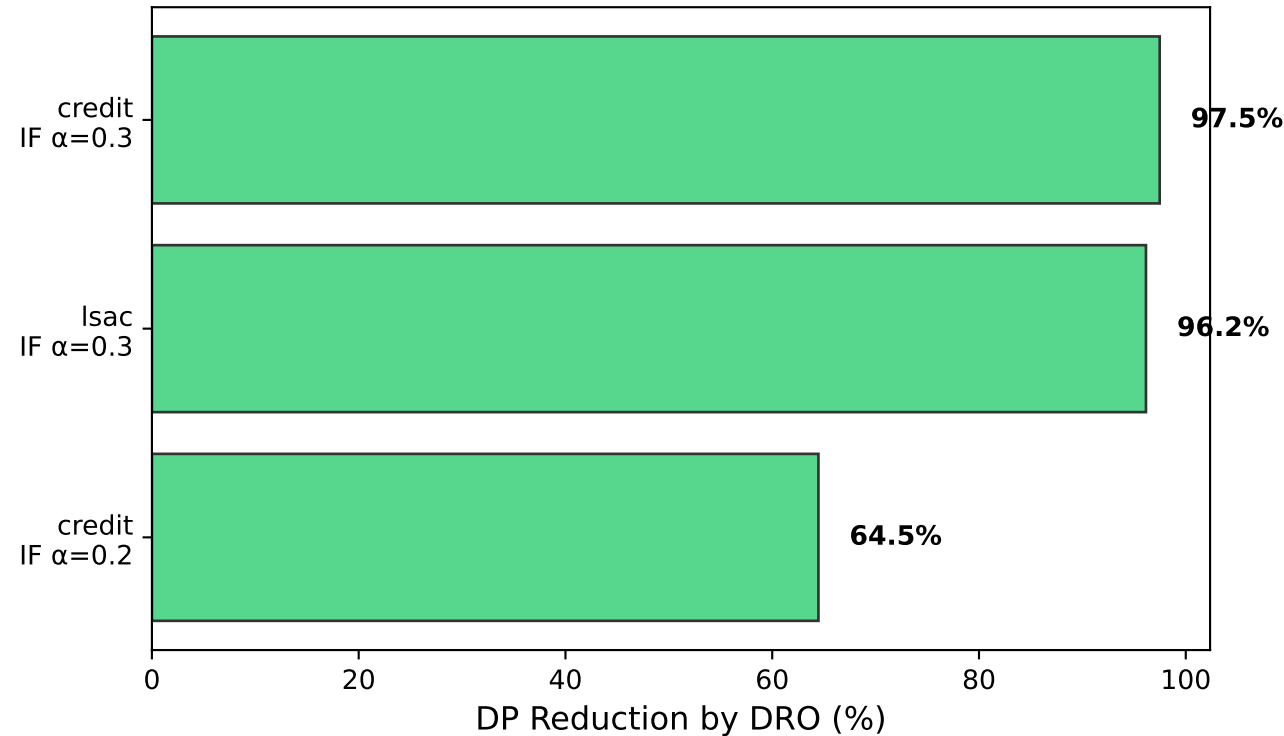
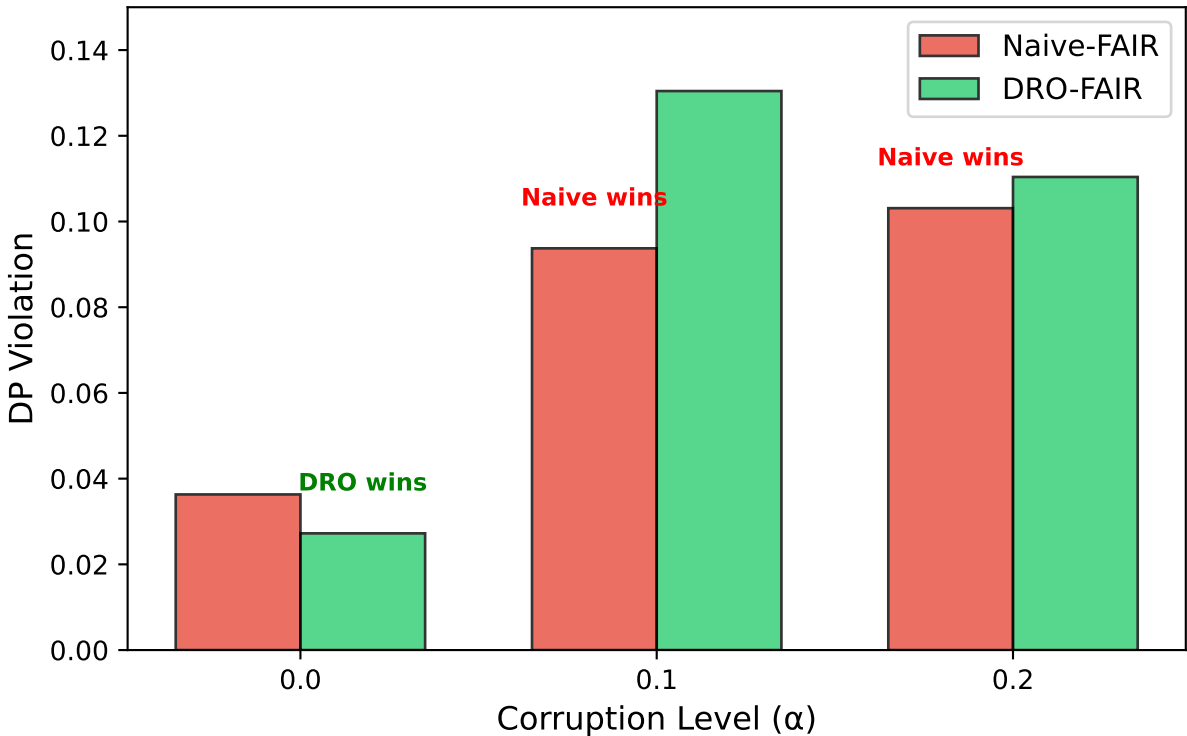


Weekly Research Summary — May 29, 2026 | Adversarial Fairness Attacks on DRO-FAIR

**Task 1: DRO Improvement Under Adversarial Attacks
(Statistically Significant Results)**



**Task 2: UTKFace Under Label Corruption
(ResNet18 Features, 23K Images)**



KEY FINDING: DRO-FAIR Robustness is NOT Universal

On TABULAR DATA (Credit, LSAC):

- DRO reduces DP violation by 64-97% under IF-targeted adversarial attacks ($p < 0.05$)
- BUT: DRO loses under DP-targeted attacks – the adversary is strong enough to break it

On IMAGE DATA (UTKFace, ResNet18 features):

- DRO makes fairness WORSE under corruption – Naive-FAIR outperforms DRO by 7-39%
- This is OPPOSITE to tabular results – a surprising, novel finding

CONCLUSION: DRO's robustness depends on BOTH the attack metric (IF vs DP) AND the data modality (tabular vs image features). Prior work only tested random noise – we show gradient-based adversarial attacks reveal important limitations.